Lesson 09: How do we generate electricity?

Name: _____

Work with a partner. Visit each place once. If a bench is busy, try another. Do the test, jot down what you see and reset the kit. Step away to talk and write.

Ngārunui Beach

Magnets are attached to the underside of the weights.

1. Keep the weights still. What does the meter read? 0.00

2. Gently push the weights down a little, then let go. What does the meter do as the weights move? moves many different smaller readings

3. Talk with your partner. Why do you think that happened?

the plants and potential energy creates a small amount of electricity that gets picked up by the ammeter

Papahua Skatepark

1. Keep the motor still. What does the meter read? 0.00

2. Spin the shaft of the motor slowly, then faster. What does the meter show?

Slow: small numbers on the ammeter Faster: larger numbers on the ammeter

3. Talk with your partner. Why do you think that happened?

the rotating of the shaft creates force that turns to electrical energy or power read by the ammeter

Manu Bay

Look inside the motor. Sketch what you see. Draw an arrow to show what turns. Then step away.

After visiting all three places

1. What two parts can you find in both the motor and the Ngārunui Beach setup?

magnets and copper coiling

2. What moved at Ngārunui Beach? the weights and arms

What turned at Papahua Skatepark? the motors shaft

3. Talk with your partner. Why do you think the meter changed in both tests?

because a diffrent amount of energy was flowing through it, and because it changed from stationary to in motion

Lesson 09: How do we generate electricity?

Name: _____

Work with a partner. Visit each place once. If a bench is busy, try another. Do the test, jot down what you see and reset the kit. Step away to talk and write.

Ngārunui Beach

Magnets are attached to the underside of the weights.

1. Keep the weights still. What does the meter read? 00.1

2. Gently push the weights down a little, then let go. What does the meter do as the weights move? 00.3

3. Talk with your partner. Why do you think that happened?

Magnetic force??? Something etc do with movement of the magnets

Papahua Skatepark

1. Keep the motor still. What does the meter read? _____

2. Spin the shaft of the motor slowly, then faster. What does the meter show?

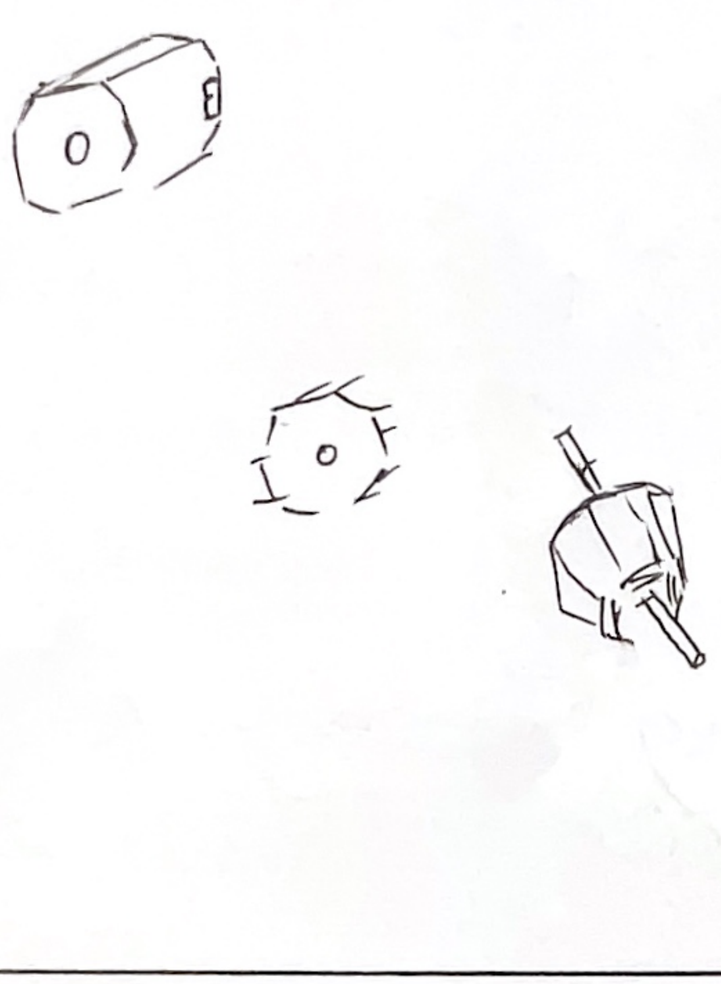
Slow: 0.1 -> 0.25 Faster: 25.9 -> 13.26

3. Talk with your partner. Why do you think that happened?

when we spin the metal around on we generated energy through the movement.

Manu Bay

Look inside the motor. Sketch what you see. Draw an arrow to show what turns. Then step away.



After visiting all three places

1. What two parts can you find in both the motor and the Ngārunui Beach setup?

Amphmeter and Spring / Magnets

2. What moved at Ngārunui Beach? The Magnets

What turned at Papahua Skatepark? Metal Spinning Rod / Shafts of the Motor.

3. Talk with your partner. Why do you think the meter changed in both tests?

Something to do whith the movment
& from the weight & wires.

Lesson 09: How do we generate electricity?

Name: _ _____

Work with a partner. Visit each place once. If a bench is busy, try another. Do the test, jot down what you see and reset the kit. Step away to talk and write.

Ngārunui Beach

Magnets are attached to the underside of the weights.

1. Keep the weights still. What does the meter read? 0.0

2. Gently push the weights down a little, then let go. What does the meter do as the weights move? moves negative then positive.

3. Talk with your partner. Why do you think that happened?

because theres energy in the glass being recorded

Papahua Skatepark

1. Keep the motor still. What does the meter read? 0.0

2. Spin the shaft of the motor slowly, then faster. What does the meter show?

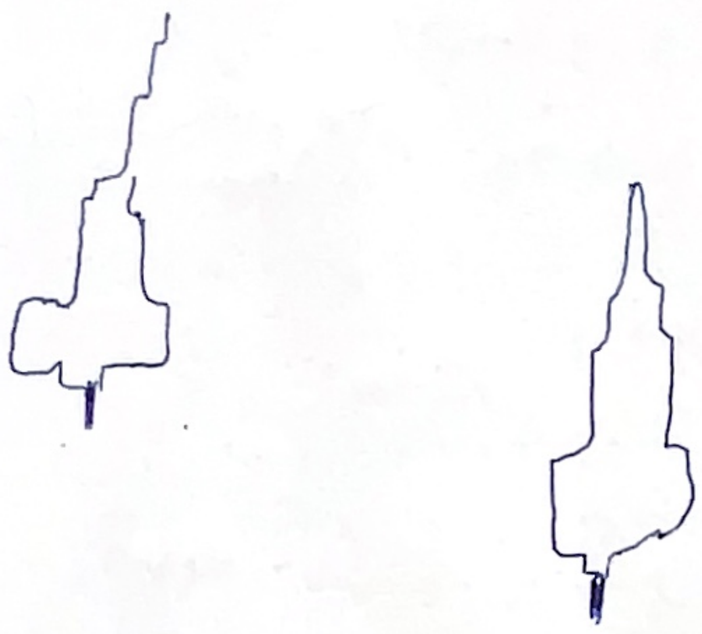
Slow: Off Low number Faster: Big numbers.

3. Talk with your partner. Why do you think that happened?

? causes friction with magnets that makes electricity

Manu Bay

Look inside the motor. Sketch what you see. Draw an arrow to show what turns. Then step away.



After visiting all three places

1. What two parts can you find in both the motor and the Ngārunui Beach setup?

magnets and wires

2. What moved at Ngārunui Beach? the weights

   What turned at Papahua Skatepark? the little motor knob.

3. Talk with your partner. Why do you think the meter changed in both tests?

"Because the magnets are causing voltage"

Lesson 09: How do we generate electricity?

Name _____

Work with a partner. Visit each place once. If a bench is busy, try another. Do the test, jot down what you see and reset the kit. Step away to talk and write.

Ngārunui Beach

Magnets are attached to the underside of the weights.

1. Keep the weights still. What does the meter read? 0.00

2. Gently push the weights down a little, then let go. What does the meter do as the weights move? Moves neg then pos

3. Talk with your partner. Why do you think that happened?

because theres energy in the glass being recoiled.

Papahua Skatepark

1. Keep the motor still. What does the meter read? 0.00

2. Spin the shaft of the motor slowly, then faster. What does the meter show?

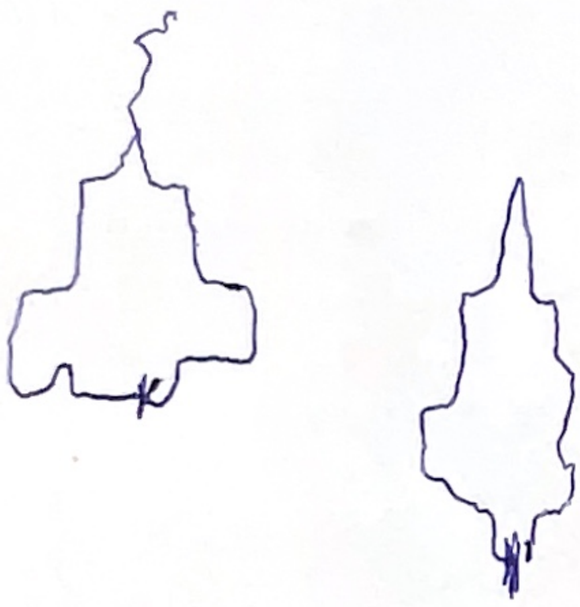
Slow: Smaller number Faster: Bigger numbers

3. Talk with your partner. Why do you think that happened?

Faster you Spin it more Kinetic energy it causes because of friction with magnets. Which causes higher volts When you Spin it fast.

Manu Bay

Look inside the motor. Sketch what you see. Draw an arrow to show what turns. Then step away.



After visiting all three places

1. What two parts can you find in both the motor and the Ngārunui Beach setup?

Magnets and Wires

2. What moved at Ngārunui Beach? the Weights

What turned at Papahua Skatepark? the little thing.

3. Talk with your partner. Why do you think the meter changed in both tests?

because the magnets are causing voltage.

Lesson 09: How do we generate electricity?

Name: _____

Work with a partner. Visit each place once. If a bench is busy, try another. Do the test, jot down what you see and reset the kit. Step away to talk and write.

Ngārunui Beach

Magnets are attached to the underside of the weights.

1. Keep the weights still. What does the meter read? 0.00

2. Gently push the weights down a little, then let go. What does the meter do as the weights move? It goes up to 5-8-5. then goes down and up =

3. Talk with your partner. Why do you think that happened?

Because the weights are still moving

Papahua Skatepark

1. Keep the motor still. What does the meter read? 00.00

2. Spin the shaft of the motor slowly, then faster. What does the meter show?

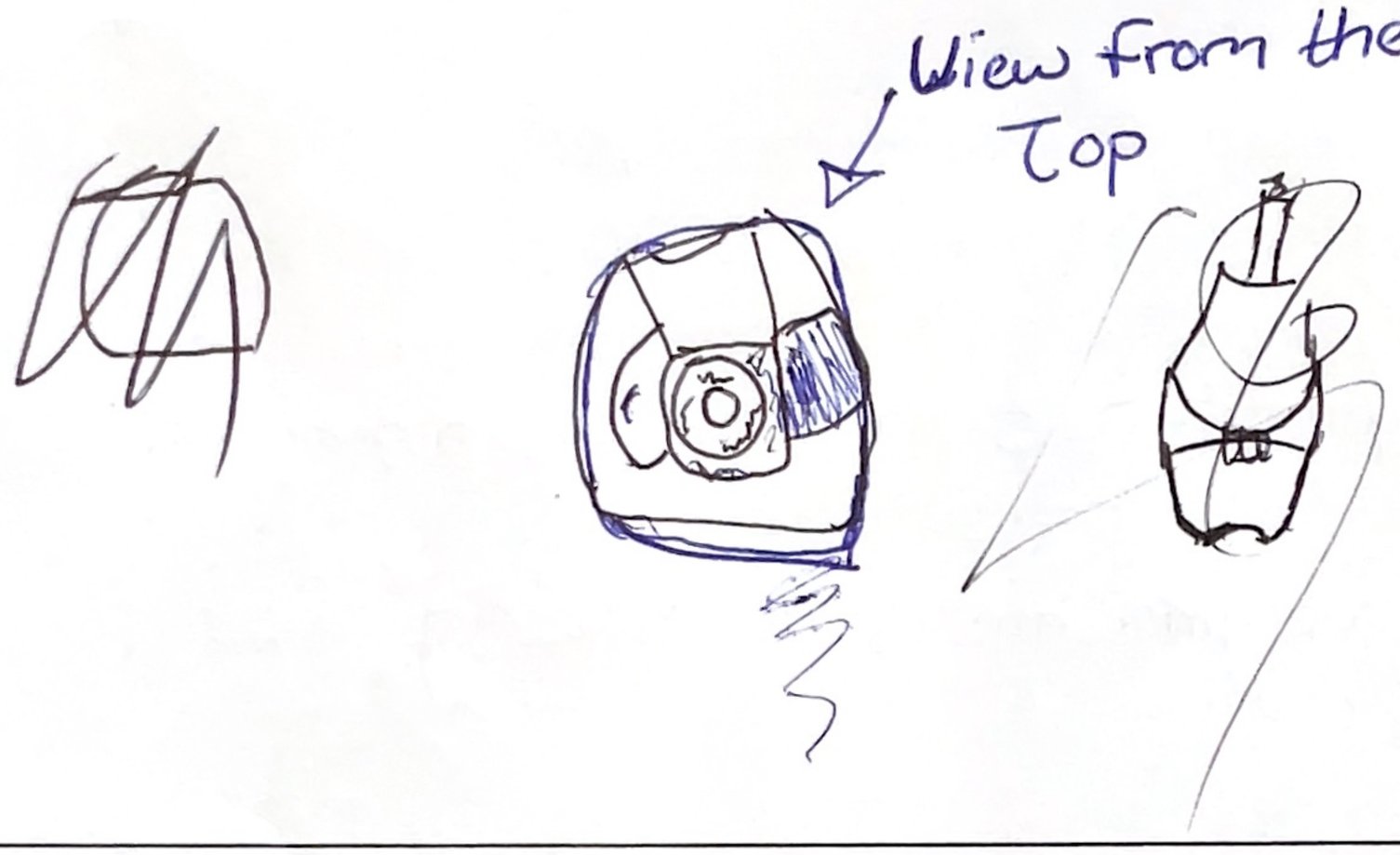
Slow: 00.6 Faster: 54

3. Talk with your partner. Why do you think that happened?

Beacuce it moving upand down

Manu Bay

Look inside the motor. Sketch what you see. Draw an arrow to show what turns. Then step away.



After visiting all three places

1. What two parts can you find in both the motor and the Ngārunui Beach setup?

an Ampmeter and magnets.

2. What moved at Ngārunui Beach? the weights

   What turned at Papahua Skatepark? the shaft motor

3. Talk with your partner. Why do you think the meter changed in both tests?

Because your moving something. So it will be forced to show.

Lesson 09: How do we generate electricity?

Name: _____

Work with a partner. Visit each place once. If a bench is busy, try another. Do the test, jot down what you see and reset the kit. Step away to talk and write.

Ngārunui Beach

Magnets are attached to the underside of the weights.

1. Keep the weights still. What does the meter read? 0000

2. Gently push the weights down a little, then let go. What does the meter do as the weights move? Jumps between 12-47 rapidly

3. Talk with your partner. Why do you think that happened?

more energy going through it

Papahua Skatepark

1. Keep the motor still. What does the meter read? 000

2. Spin the shaft of the motor slowly, then faster. What does the meter show?

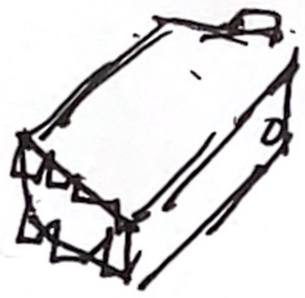
Slow: 1.3 ~ 1.6 Faster: ~~128~~ Caverage 12.8 ~ 91.1

3. Talk with your partner. Why do you think that happened?

moving it generates energy

Manu Bay

Look inside the motor. Sketch what you see. Draw an arrow to show what turns. Then step away.



After visiting all three places

1. What two parts can you find in both the motor and the Ngārunui Beach setup?

Magnets and ~~Coil~~ Ammeter

2. What moved at Ngārunui Beach? the weight

What turned at Papahua Skatepark? the motor crank

3. Talk with your partner. Why do you think the meter changed in both tests?

Because of the energy levels (more electrical energy, changing voltage)